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Signal processing system and signal processing method where a protocol analysis circuit is used to simplify hardware structure and reduce power consumption

Abstract

A signal processing system includes a first transceiver unit, a second transceiver unit, a protocol analysis circuit, a system chip, and a network unit. The first transceiver unit can transceive a first optical signal and a first electrical signal. The second transceiver unit can transceive a second optical signal and a second electrical signal. The protocol analysis circuit can process the first electrical signal and an analysis signal related to the first optical signal. The system chip can process the analysis signal, the second electrical signal, a first operation signal and a second operation signal. The network unit can transceive the first operation signal and the second operation signal, and transceive a first network signal and a second network signal between the network unit and a user device. The system chip and the network unit can process signals related to the first optical signal and the second optical signal.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/422,437, filed on Nov. 4, 2022. The content of the application is incorporated herein by reference.

BACKGROUND OF THE INVENTION
1. Field of the Invention
(1) The application is related to a signal processing system and signal processing method, and more particularly, a signal processing system and signal processing method where a protocol analysis circuit is used to simplify hardware structure and reduce power consumption.
2. Description of the Prior Art
(2) As the demand for high-speed data transmissions increases, communication technology has also developed accordingly. At present, a passive optical network (PON) can be used for data

transmissions through optical fibers. For example, a gigabit passive optical network (GPON) can be used in the communication field for high-speed data transmissions.

(3) However, because of the limitations of related hardware and software, when the data transmissions of two passive optical networks of two different speeds are needed for an application, two sets of different hardware must be used. For example, if a user needs to perform data transmissions of 2.5 gigabits per second (2.5 Gbps) and 10 gigabits per second (10 Gbps), data transmissions of 2.5 gigabits per second (2.5 Gbps) need dedicated transceiver unit, laser driver, system on chip, memory and network unit; and data transmissions of 10 gigabits per second (10 Gbps) need another set of dedicated transceiver unit, laser driver, system on chip, memory and network unit. Therefore, it is difficult to reduce the complexity and power consumption of the system. In the field, a suitable solution is in need to simplify hardware requirements.

SUMMARY OF THE INVENTION

(4) An embodiment provides a signal processing system including a first transceiver unit, a second transceiver unit, a protocol analysis circuit, a system chip and a network unit. The first transceiver unit is configured to transceive a first optical signal and a first electrical signal. The second transceiver unit is configured to transceive a second optical signal and a second electrical signal. The protocol analysis circuit is coupled to the first transceiver unit, and configured to process the first electrical signal and an analysis signal and transceive the first electrical signal and the analysis signal. The system chip is coupled to the protocol analysis circuit and the second transceiver unit, and configured to process the analysis signal, the second electrical signal, a first operation signal and a second operation signal, and transceive the analysis signal, the second electrical signal, the first operation signal and the second operation signal. The network unit is coupled to the system chip and a user device, and configured to transceive the first operation signal and the second operation signal, and transceive a first network signal and a second network signal between the network unit and the user device. The first electrical signal, the analysis signal, the first operation signal and the first network signal are corresponding to the first optical signal. The second electrical signal, the second operation signal and the second network signal are corresponding to the second optical signal.

(5) Another embodiment provides a signal processing method for a signal processing system. The signal processing system can include a first transceiver unit, a second transceiver unit, a protocol analysis circuit, a system chip and a network unit. The signal processing method can include using the first transceiver unit to transceive a first optical signal and a first electrical signal corresponding to the first optical signal, using the second transceiver unit to transceive a second optical signal and a second electrical signal corresponding to the second optical signal, using the protocol analysis circuit to transceive the first electrical signal and an analysis signal corresponding to the first electrical signal, and process the first electrical signal and the analysis signal, using the system chip to transceive the analysis signal, a first operation signal corresponding to the analysis signal, the second electrical signal, and a second operation signal corresponding to the second electrical signal, and process the analysis signal, the first operation signal, the second electrical signal and the second operation signal, and using the network unit to transceive the first operation signal, the second operation signal, a first network signal corresponding to the first operation signal, and a second network signal corresponding to the second operation signal.

(6) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** illustrates a signal processing system according to an embodiment.

(2) FIG. **2** illustrates a path of processing the signals related to the first optical signal in FIG. **1** according to an embodiment.

(3) FIG. **3** illustrates another path of processing the signals related to the second optical signal in FIG. **1** according to an embodiment.

(4) FIG. **4** illustrates the operations of the protocol analysis circuit in FIG. **1** according to an embodiment.

(5) FIG. **5** illustrates the operations of the system chip and the memory in FIG. **1** according to an embodiment.

(6) FIG. **6** illustrates a flow chart of a signal processing method for the signal processing system in FIG. **1** according to an embodiment.

DETAILED DESCRIPTION

(7) FIG. **1** illustrates a signal processing system **100** according to an embodiment. The signal processing system **100** can be used to transceive signals of two different bit rates, as described below.

(8) The signal processing system **100** can include a first transceiver unit **110**, a second transceiver unit **120**, a protocol analysis circuit **130**, a system chip **140**, and a network unit **150**.

(9) The first transceiver unit **110** can be used to transceive a first optical signal S**1** and a first electrical signal S**1***a*. The second transceiver unit **120** can be used to transceive a second optical signal S**2** and a second electrical signal S**2***a*. The protocol analysis circuit **130** can be coupled to the first transceiver unit **110** and used to transceive and process the first electrical signal S**1***a* and an analysis signal S**1***b*. The system chip **140** can be coupled to the protocol analysis circuit **130** and the second transceiver unit **120** and used to transceive and process the analysis signal S**1***b*, the second electrical signal S**2***a*, a first operation signal S**1***c* and a second operation signal S**2***c*. According to embodiments, the system chip **140** can include a system on chip (SoC) and/or a processor. The network unit **150** can be coupled to the system chip **140** and a user device **188** and used to transceive the first operation signal S**1***c* and the second operation signal S**2***c* and transceive a first network signal S**1***d* and a second network signal S**2***d* between the network unit **150** and the user device **188**. According to embodiments, the network unit **150** can be an Ethernet unit.

(10) In FIG. **1**, the first electrical signal S**1***a*, the analysis signal S**1***b*, the first operation signal S**1***c* and the first network signal S**1***d* can be corresponding to the first optical signal S**1**; and the second electrical signal S**2***a*, the second operation signal S**2***c* and the second network signal S**2***d* can be corresponding to the second optical signal S**2**, as shown in Table 1.

(11) TABLE-US-00001 TABLE 1 Signals related to the first The first electrical signal Sla optical signal S1 The analysis signal S1b The first operation signal S1c The first network signal S1d Signals related to the second The second electrical signal S2a optical signal S2 The second operation signal S2c The second network signal S2d

(12) The first optical signal S**1** can be corresponding to a first bit rate, and the second optical signal S**2** can be corresponding to a second bit rate lower than the first bit rate. For example, the first bit rate can be 10 Gigabits per second (Gbps), and the second bit rate can be 2.5 Gbps. This is an example, and embodiments are not limited thereto.

(13) As shown in FIG. **1**, the first transceiver unit **110** can include a first optical assembly unit **112** and a first driver **114**. The first optical assembly unit **112** can be used to transceive the first optical signal S**1**. The first driver **114** can be coupled to the first optical assembly unit **112**, and used to convert the first optical signal S**1** to the first electrical signal S**1***a* and convert the first electrical signal S**1***a* to the first optical signal S**1**. The second transceiver unit **120** can include a second optical assembly unit **122** and a second driver **124**. The second optical assembly unit **122** can be used to transceive the second optical signal S**2**. The second driver **124** can be coupled to the second

optical assembly unit **122**, and used to convert the second optical signal S**2** to the second electrical signal S**2***a* and convert the second electrical signal S**2***a* to the second optical signal S**2**. For example, each of the first optical assembly unit **112** and the second optical assembly unit **122** can include a bi-directional optical sub-assembly (BOSA). For example, each of the first driver **114** and the second driver **124** can include a laser driver used to perform conversion between an optical signal and an electrical signal.

(14) As shown in FIG. **1**, the signal processing system **100** can further include a memory **160** coupled to the system chip **140** and used to perform a queue operation related to the second electrical signal S**2***a* and the second operation signal S**2***c*. The queue operation of the memory **160** is not related to the first optical signal S**1** and the first electrical signal S**1***a*. In other words, when the signal processing system **100** is used to process the first optical signal S**1** and the signals related to the first optical signal S**1**, the memory **160** is not used, and more details are described below. According to embodiments, the memory **160** can include a dynamic random-access memory (DRAM). For example, the memory **160** can include a double data rate (DDR) memory.

(15) As shown in FIG. **1**, the signal processing system **100** can further include a memory **165** coupled to the system chip **140**. For example, the memory **165** can include a flash memory used to store programs and parameters related to a boot process.

(16) According to embodiments, the signal processing system **100** can further include a first optical network unit (ONU) interface **192**, a second optical network unit interface **194**, a port physical layer (PHY) interface **196**, a first network interface **198** and a second network interface **199**.

(17) As shown in FIG. **1**, the first optical network unit interface **192** can be disposed between the first transceiver unit **110** and the protocol analysis circuit **130** for transceiving the first electrical signal S**1***a*. The second optical network unit interface **194** can be disposed between the second transceiver unit **120** and the system chip **140** for transceiving the second electrical signal S**2***a*. The port physical layer interface **196** can be disposed between the protocol analysis circuit **130** and the system chip **140** for transceiving the analysis signal S**1***b*. The first network interface **198** can be disposed between the system chip **140** and the network unit **150** for transceiving the first operation signal S**1***c* and the second operation signal S**2***c*. The second network interface **199** can be disposed between the network unit **150** and the user device **188** for transceiving the first network signal S**1***d* and the second network signal S**2***d*. For example, the bit rate of the first network interface **198** can be 10 Gbps, and the first network interface **198** can be an XFI interface (e.g. Ziffy interface). For example, the second network interface **199** can be an RJ45 interface.

(18) The signal processing system **100** can further include a first memory interface **193** and a second memory interface **195**. The first memory interface **193** can be disposed between the system chip **140** and the memory **160**. The second memory interface **195** can be disposed between the system chip **140** and the memory **165**. For example, the first memory interface **193** can be a double data rate (DDR) interface, and the second memory interface **195** can be a serial peripheral interface (SPI).

(19) As shown in FIG. **1**, the first optical signal S**1** can include a signal S**11** and a signal S**12**. The signal S**11** can be transmitted from outside of the signal processing system **100** to the first transceiver unit **110**. The signal S**12** can be transmitted from the first transceiver unit **110** to outside of the signal processing system **100**. In other words, the first transceiver unit **110** can be used to receive the signal S**11** or transmit the signal S**12**.

(20) The second optical signal S**2** can include a signal S**21** and a signal S**22**. The signal S**21** can be transmitted from outside of the signal processing system **100** to the second transceiver unit **120**. The signal S**22** can be transmitted from the second transceiver unit **120** to outside of the signal processing system **100**. In other words, the second transceiver unit **120** can be used to receive the signal S**21** or transmit the signal S**22**.

(21) The first electrical signal S**1***a* can include a signal S**1***a***1** and a signal S**1***a***2**. The signal S**1***a***1** can be transmitted from the first transceiver unit **110** to the protocol analysis circuit **130**. The signal

S**1***a***2** can be transmitted from the protocol analysis circuit **130** to the first transceiver unit **110**.
(22) The second electrical signal S**2***a* can include a signal S**2***a***1** and a signal S**2***a***2**. The signal S**2***a***1** can be transmitted from the second transceiver unit **120** to the system chip **140**. The signal S**2***a***2** can be transmitted from the system chip **140** to the second transceiver unit **120**.
(23) The analysis signal S**1***b* can include a signal S**1***b***1** and a signal S**1***b***2**. The signal S**1***b***1** can be transmitted from the protocol analysis circuit **130** to the system chip **140**. The signal S**1***b***2** can be transmitted from the system chip **140** to the protocol analysis circuit **130**.
(24) The first operation signal S**1***c* can include a signal S**1***c***1** and a signal S**1***c***2**. The signal S**1***c***1** can be transmitted from the system chip **140** to the network unit **150**. The signal S**1***c***2** can be transmitted from the network unit **150** to the system chip **140**.
(25) The second operation signal S**2***c* can include a signal S**2***c***1** and a signal S**2***c***2**. The signal S**2***c***1** can be transmitted from the system chip **140** to the network unit **150**. The signal S**2***c***2** can be transmitted from the network unit **150** to the system chip **140**.
(26) The first network signal S**1***d* can include a signal S**1***d***1** and a second S**1***d***2**. The signal S**1***d***1** can be transmitted from the network unit **150** to the user device **188**. The signal S**1***d***2** can be transmitted from the user device **188** to the network unit **150**.
(27) The second network signal S**2***d* can include a signal S**2***d***1** and a second S**2***d***2**. The signal S**2***d***1** can be transmitted from the network unit **150** to the user device **188**. The signal S**2***d***2** can be transmitted from the user device **188** to the network unit **150**.
(28) FIG. **2** illustrates a path PT**1** of processing the signals related to the first optical signal S**1** in FIG. **1** according to an embodiment.
(29) As shown by the path PT**1**, when the signal processing system **100** receives the first optical signal S**1**, the first transceiver unit **110** can convert the first optical signal S**1** to the first electrical signal S**1***a*. The protocol analysis circuit **130** can generate the analysis signal S**1***b* according to the first electrical signal S**1***a*. The system chip **140** can generate the first operation signal S**1***c* according to the analysis signal S**1***b*. The network unit **150** can generate the first network signal S**1***d* according to the first operation signal S**1***c* and transmit the first network signal S**1***d* to the user device **188**.
(30) In FIG. **2**, the signal S**1***a***1** can be generated according to the signal S**11**. The signal S**1***b***1** can be generated according to the signal S**1***a***1**. The signal S**1***c***1** can be generated according to the signal S**1***b***1**. The signal S**1***d***1** can be generated according to the signal S**1***c***1**. In detail, when the first transceiver unit **110** receives the signal S**11**, the signal S**1***a***1** can be generated according to the signal S**11**, and the signal S**1***a***1** can be transmitted to the protocol analysis circuit **130** through the first optical network unit interface **192**. When the protocol analysis circuit **130** receives the signal S**1***a***1**, the signal S**1***b***1** can be generated according to the signal S**1***a***1**, and the signal S**1***b***1** can be transmitted to the system chip **140** through the port physical layer interface **196**. When the system chip **140** receives the signal S**1***b***1**, the signal S**1***c***1** can be generated according to the signal S**1***b***1**, and the signal S**1***c***1** can be transmitted to the network unit **150** through the first network interface **198**. When the network unit **150** receives the signal S**1***c***1**, the signal S**1***d***1** can be generated according to the signal S**1***c***1**, and the signal S**1***d***1** can be transmitted to the user device **188** through the second network interface **199**.
(31) As shown by the path PT**1**, when the network unit **150** receives the first network signal S**1***d* from the user device **188**, the network unit **150** can generate the first operation signal S**1***c* according to the first network signal S**1***d*. The system chip **140** can generate the analysis signal S**1***b* according to the first operation signal S**1***c*. The protocol analysis circuit can generate the first electrical signal S**1***a* according to the analysis signal S**1***b*. The first transceiver unit **110** can covert the first electrical signal S**1***a* to the first optical signal S**1**.
(32) In FIG. **2**, the signal S**1***c***2** can be generated according to the signal S**1***d***2**. The signal S**1***b***2** can be generated according to the signal S**1***c***2**. The signal S**1***a***2** can be generated according to the signal S**1***b***2**. The signal S**12** can be generated according to the signal S**1***a***2**. In detail, the user

device **188** can transmit the signal S**1***d***2** to the network unit **150** through the second network interface **199**. The network **150** can receive the signal S**1***d***2**, generate the signal S**1***c***2** according to the signal S**1***d***2**, and transmit the signal S**1***c***2** to the system chip **140** through the first network interface **198**. The system chip **140** can receive the signal S**1***c***2**, generate the signal S**1***b***2** according to the signal S**1***c***2**, and transmit the signal S**1***b***2** to the protocol analysis circuit **130** through the port physical layer interface **196**. The protocol analysis circuit **130** can receive the signal S**1***b***2**, generate the signal S**1***a***2** according to the signal S**1***b***2**, and transmit the signal S**1***a***2** to the first transceiver unit **110** through the first optical network unit interface **192**. The first transceiver unit **110** can receive the signal S**1***a***2**, generate the signal S**12** according to the signal S**1***a***2**, and transmit the signal S**12**.
(33) As shown by the path PT**1**, when the signal processing system **100** is used to process the signals related to the first optical signal S**1**, the protocol analysis circuit **130** can be used to perform encryption, decryption and classification-and-forwarding without using the memory **160** to perform a queue operation.
(34) FIG. **3** illustrates a path PT**2** of processing the signals related to the second optical signal S**2** in FIG. **1** according to an embodiment.
(35) As shown by the path PT**2**, when the signal processing system **100** receives the second optical signal S**2**, the second transceiver unit **120** can convert the second optical signal S**2** to the second electrical signal S**2***a*. The system chip **140** can receive the second electrical signal S**2***a* to process the second electrical signal S**2***a*. The memory **160** can perform a queue operation for the packets and tasks related to the second electrical signal S**2***a*. After the system chip **140** processes the second electrical signal S**2***a*, the system chip **140** can generate the second operation signal S**2***c*. The network unit **150** can generate the second network signal S**2***d* according to the second operation signal S**2***c*, and transmit the second network signal S**2***d* to the user device **188**.
(36) In FIG. **3**, the signal S**2***a***1** can be generated according to the signal S**21**, the signal S**2***c***1** can be generated according to the signal S**2***a***1**, and the signal S**2***d***1** can be generated according to the signal S**2***c***1**.
(37) In detail, the second transceiver unit **120** can receive the signal S**21**, generate the signal S**2***a***1** according to the signal S**21**, and transmit the signal S**2***a***1** to the system chip **140** through the second optical network unit interface **194**. The system chip **140** can receive the signal S**2***a***1**, process the signal S**2***a***1**, use the memory **160** to perform a queue operation, generate the signal S**2***c***1** according to the signal S**2***a***1**, and transmit the signal S**2***c***1** to the network unit **150** through the first network interface **198**. The network unit **150** can receive the signal S**2***c***1**, generate the signal S**2***d***1** according to the signal S**2***c***1**, and transmit the signal S**2***d***1** to the user device **188** through the second network interface **199**.
(38) As shown by the path PT**2**, when the network unit **150** receives the second network signal S**2***d* from the user device **188**, the network device **150** can generate the second operation signal S**2***c* according to the second network signal S**2***d*. The system chip **140** can receive the second operation signal S**2***c*, process the second operation signal S**2***c*, and use the memory **160** to perform a queue operation for the packets and tasks related to the second operation signal S**2***c*. After the system chip **140** processes the second operation signal S**2***c*, the system chip **140** can generate the second electrical signal S**2***a* accordingly. Then, the second transceiver unit **120** can convert the second electrical signal S**2***a* to the second optical signal S**2**.
(39) In FIG. **3**, the signal S**2***c***2** can be generated according to the signal S**2***d***2**, the signal S**2***a***2** can be generated according to the signal S**2***c***2**, and the signal S**22** can be generated according to the signal S**2***a***2**.
(40) In detail, the user device **188** can transmit the signal S**2***d***2** to the network unit **150** through the second network interface **199**. After the network unit **150** receives the signal S**2***d***2**, the network unit **150** can generate the signal S**2***c***2** according to the signal S**2***d***2**, and transmit the signal S**2***c***2** to the system chip **140** through the first network interface **198**. After the system chip **140** receives the

signal S**2***c***2**, the system chip **140** can process the signal S**2***c***2**, use the memory **160** to perform a queue operation, generate the signal S**2***a***2** according to the signal S**2***c***2**, and transmit the signal S**2***a***2** to the second transceiver unit **120** through the second optical network unit interface **194**. After the second transceiver unit **120** receives the signal S**2***a***2**, the second transceiver unit **120** can generate the signal S**22** according to the signal S**2***a***2**, and transmit the signal S**22**.
(41) As shown by the path PT**2**, when the signal processing system **100** is used to process the signals related to the second optical signal S**2**, the protocol analysis circuit **130** is not used, and the memory **160** is used to perform the queue operation.
(42) FIG. **4** illustrates the operations of the protocol analysis circuit **130** in FIG. **1** according to an embodiment. The protocol analysis circuit **130** can include a host **132** and a network processor **134**. The network processor **134** can include a classification-and-forwarding unit **1342** and an area network unit **1344**. When an internet protocol packet (TP packet, referred to as packet in the text) carried by an electrical signal reaches the protocol analysis circuit **130**, the packet can be transmitted to the network processor **134** through the host **132**. According to the descriptions of the low-level assembly language in the packet, the packet can be transmitted to the classification-and-forwarding unit **1342** or the area network unit **1344**.
(43) If the packet is transmitted to the classification-and-forwarding unit **1342**, the protocol analysis circuit **130** can perform a classification-and-forwarding operation, arrange the packet in a queue, and then transmit the packet through the area network unit **1344**. As shown in FIG. **4**, the corresponding low-level assembly language may be rdpa_cpu_send_xxx( ) Method=Bridge. This is an example, and embodiments are not limited thereto.
(44) If the packet is not transmitted to the classification-and-forwarding unit **1342**, the protocol analysis circuit **130** can arrange the packet in a queue, and then transmit the packet through the area network unit **1344**. As shown in FIG. **4**, the corresponding low-level assembly language may be rdpa_cpu_send_xxx( ). This is an example, and embodiments are not limited thereto.
(45) The abovementioned queue can be a random queue. The area network unit **1344** can assign ports of a wide area network (WAN) and/or a local area network (LAN) for transmitting packets.
(46) In FIG. **4**, the classification-and-forwarding unit **1342** and the area network unit **1344** of the network processor **134** can be implemented using appropriate hardware and/or software. The classification-and-forwarding unit **1342** and the area network unit **1344** can be separated as different units or integrated as one unit.
(47) As shown in FIG. **2** and FIG. **4**, the protocol analysis circuit **130** can be used to perform an encryption operation and a decryption operation to process the analysis signal S**1***b* and the first electrical signal S**1***a* corresponding to the first optical signal S**1**. The protocol analysis circuit **130** can be used to perform a classification-and-forwarding operation to process the analysis signal S**1***b* and the first electrical signal S**1***a* corresponding to the first optical signal S**1**.
(48) FIG. **5** illustrates the operations of the system chip **140** and the memory **160** in FIG. **1** according to an embodiment. As shown in FIG. **5**, the system chip **140** can include a host **142** and a network processor **144**. The network processor **144** can include an area network unit **1442**, a reason-to-queue mapping unit **1444** and a direct memory access (DMA) unit **1446**. When a packet arrives the system chip **140**, the packet can be accessed through the area network unit **1442**, and then the packet can be transmitted to the reason-to-queue mapping unit **1444**. The area network unit **1442** can access information of ports of a wide area network (WAN) and/or a local area network (LAN) for transmitting the packet. The reason-to-queue mapping unit **1444** can determine whether the packet is transmitted to the direct memory access unit **1446** or the host **142**.
(49) If the packet is transmitted to the host **142**, the packet is processed without using the memory **160**. For example, in FIG. **2**, the packet related to the first optical signal S**1** is processed without using the memory **160**.
(50) If the packet is transmitted to the direct memory access (DMA) unit **1446**, the memory **160** can be used to perform a queue operation to arrange and schedule the packet and related tasks. For

example, the queue operation of the memory **160** can be a ring queue for scheduling and buffering related data. After performing the queue operation in the memory **160**, the packet can be transmitted to the host **142** of the system chip **140** so as to transmit the packet from the system chip **140** to outside of the system chip **140**. For example, the low-level assembly language corresponding to the operation can be rdpa_cpu_packet_get( ). This is an example, and embodiments are not limited thereto. As shown in FIG. **3** and FIG. **5**, the packets related to the second optical signal S**2** can be processed using the memory **160**.

(51) In FIG. **5**, the area network unit **1442**, the reason-to-queue mapping unit **1444** and the direct memory access unit **1446** can be implemented using appropriate hardware and/or software. The area network unit **1442**, the reason-to-queue mapping unit **1444** and the direct memory access unit **1446** can be separated as different units or integrated as one unit.

(52) As mentioned above, the queue operation performed with the memory **160** is related to the second optical signal S**2**, the second electrical signal S**2***a* and the second operation signal S**2***c*. The queue operation performed with the memory **160** is not related to the first optical signal S**1**, the first electrical signal S**1***a* and the first operation signal S**1***c*.

(53) FIG. **6** illustrates a flow chart of a signal processing method **600** for the signal processing system **100** according to an embodiment. As shown in FIG. **1** and FIG. **6**, the signal processing method **600** can include following steps.

(54) Step **610**: use the first transceiver unit **110** to transceive the first optical signal S**1** and the first electrical signal S**1***a* corresponding to the first optical signal S**1**;

(55) Step **620**: use the second transceiver unit **120** to transceive the second optical signal S**2** and the second electrical signal S**2***a* corresponding to the second optical signal S**2**;

(56) Step **630**: use the protocol analysis circuit **130** to transceive and process the first electrical signal S**1***a* and the analysis signal S**1***b* corresponding to the first electrical signal S**1***a;*

(57) Step **640**: use the system chip **140** to transceive and process the analysis signal S**1***b*, the first operation signal S**1***c* corresponding to the analysis signal S**1***b*, the second electrical signal S**2***a*, and the second operation signal S**2***c* corresponding to the second electrical signal S**2***a*; and

(58) Step **650**: use the network unit **150** to transceive the first operation signal S**1***c*, the second operation signal S**2***c*, the first network signal S**1***d* corresponding to the first operation signal S**1***c*, and the second network signal S**2***d* corresponding to the second operation signal S**2***c*.

(59) In FIG. **6**, the sequence and the content of the steps can be flexibly adjusted according to requirements. For example, Step **610** can be performed before or after Step **620**, or Steps **610** and **620** can be performed concurrently. For example, if the signal processing system **100** receives the first network signal S**1***d* and/or the second network signal S**2***d* to generate at least one of the first optical signal S**1** and the second optical signal S**2**, Step **650** can be performed first. For example, if the signal processing system **100** does not receive a signal related to the first optical signal S**1**, the steps related to the first optical signal S**1** may be omitted selectively. For example, if the signal processing system **100** does not receive a signal related to the second optical signal S**2**, the steps related to the second optical signal S**2** may be omitted selectively.

(60) In summary, in the signal processing system **100**, the circuits related to the analysis signal S**1***b* and the first electrical signal S**1***a* corresponding to the first optical signal S**1** can be embedded in the protocol analysis circuit **130**, so the system chip **140** and the network unit **150** can be shared to process the signals related to the first optical signal S**1** and the second optical signal S**2**. Hence, it is unnecessary to use two system chips, two network units and two sets of memories to process signals of two different bit rates. By using the protocol analysis circuit **130**, the first optical signal S**1** of a higher bit rate (e.g. 10 Gbps) can be processed without using the memory **160** to perform the queue operation, thus reducing the memory requirement. In addition, since there is no need to use two system chips, two network units and two sets of memories to process signals of two different bit rates, power consumption and device size can be reduced. For example, if the signal processing system **100** is not used, the power consumption of a system processing signals of two

different bit rates should be at least 200% of the power consumption of a system processing signals of only one bit rate. However, by using the signal processing system **100** and the signal processing method **600**, the power consumption can be increased by only 4%. For example, the power consumption of processing signals related to one bit rate can be 10 watts, and the power consumption of processing signals related to two different bit rates can be 10.4 watts rather than 20 watts by using the signal processing system **100** and the signal processing method **600**. As a result, the signal processing system **100** and the signal processing method **600** are useful for simplifying hardware structure and reducing power consumption and device size.
(61) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A signal processing system comprising: a first transceiver unit configured to transceive a first optical signal and a first electrical signal; a second transceiver unit configured to transceive a second optical signal and a second electrical signal; a protocol analysis circuit coupled to the first transceiver unit, and configured to process the first electrical signal and an analysis signal and transceive the first electrical signal and the analysis signal; a system chip coupled to the protocol analysis circuit and the second transceiver unit, and configured to process the analysis signal, the second electrical signal, a first operation signal and a second operation signal, and transceive the analysis signal, the second electrical signal, the first operation signal and the second operation signal; and a network unit coupled to the system chip and a user device, and configured to transceive the first operation signal and the second operation signal, and transceive a first network signal and a second network signal between the network unit and the user device; wherein the first electrical signal, the analysis signal, the first operation signal and the first network signal are corresponding to the first optical signal, and the second electrical signal, the second operation signal and the second network signal are corresponding to the second optical signal.
2. The signal processing system of claim 1, wherein: the first transceiver unit comprises: a first optical assembly unit configured to transceive the first optical signal; and a first driver coupled to the first optical assembly unit, and configured to convert the first optical signal to the first electrical signal and convert the first electrical signal to the first optical signal; and the second transceiver unit comprises: a second optical assembly unit configured to transceive the second optical signal; and a second driver coupled to the second optical assembly unit, and configured to convert the second optical signal to the second electrical signal and convert the second electrical signal to the second optical signal.
3. The signal processing system of claim 1, further comprising: a memory coupled to the system chip and configured to perform a queue operation related to the second electrical signal and the second operation signal.
4. The signal processing system of claim 3, wherein the memory is a dynamic random access memory.
5. The signal processing system of claim 1, wherein the first optical signal is corresponding to a first bit rate, and the second optical signal is corresponding to a second bit rate lower than the first bit rate.
6. The signal processing system of claim 1, wherein the protocol analysis circuit is configured to perform an encryption operation and a decryption operation to process the analysis signal and the first electrical signal.
7. The signal processing system of claim 1, wherein the protocol analysis circuit is configured to perform a classification-and-forwarding operation to process the analysis signal and the first

electrical signal.
8. The signal processing system of claim 1, further comprising: a first optical network unit interface disposed between the first transceiver unit and the protocol analysis circuit; a second optical network unit interface disposed between the second transceiver unit and the system chip; a port physical layer interface disposed between the protocol analysis circuit and the system chip; a first network interface disposed between the system chip and the network unit; and a second network interface disposed between the network unit and the user device.
9. The signal processing system of claim 1, wherein the network unit is an Ethernet unit.
10. A signal processing method for a signal processing system, the signal processing system comprising a first transceiver unit, a second transceiver unit, a protocol analysis circuit, a system chip and a network unit, the signal processing method comprising: using the first transceiver unit to transceive a first optical signal and a first electrical signal corresponding to the first optical signal; using the second transceiver unit to transceive a second optical signal and a second electrical signal corresponding to the second optical signal; using the protocol analysis circuit to transceive the first electrical signal and an analysis signal corresponding to the first electrical signal, and process the first electrical signal and the analysis signal; using the system chip to transceive the analysis signal, a first operation signal corresponding to the analysis signal, the second electrical signal, and a second operation signal corresponding to the second electrical signal, and process the analysis signal, the first operation signal, the second electrical signal and the second operation signal; and using the network unit to transceive the first operation signal, the second operation signal, a first network signal corresponding to the first operation signal, and a second network signal corresponding to the second operation signal.
11. The signal processing method of claim 10, wherein the first optical signal is corresponding to a first bit rate, and the second optical signal is corresponding to a second bit rate lower than the first bit rate.
12. The signal processing method of claim 10, wherein: the protocol analysis circuit is configured to perform an encryption operation and a decryption operation to process the analysis signal and the first electrical signal.
13. The signal processing method of claim 10, wherein: the protocol analysis circuit is configured to perform a classification-and-forwarding operation to process the analysis signal and the first electrical signal.
14. The signal processing method of claim 10, wherein: the signal processing system further comprises a memory; and the signal processing method further comprises using the memory to perform a queue operation related to the second electrical signal and the second operation signal.